Polylogarithmic Approximation for Covering and Connecting Multi-Interface Networks

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Abstract

We study problems related to connecting multi-interface networks of wireless devices. These problems can be modeled using graphs, where vertices represent the devices and edges represent potential communication links. Each vertex can activate multiple interfaces, and a connection between two vertices is established if they share at least one common active interface. We consider the heterogeneous setting, where activating an interface induces a cost that depends on the type of interface and on the vertex that activates it. In the COVERAGE problem, every connection defined in the network must be established, while in the CONNECTIVITY problem, groups of terminals specified in the input should be connected. The goal is to accomplish the required goal while minimizing either the maximum cost incurred by a vertex or the total cost incurred by all vertices. In this work we are interested in approximating the former of the two cost criterions.

To achieve this goal we model both of the problems using Integer Linear Programming (ILP) and we design approximation algorithms based on a randomized rounding of the solution of the linear programming relaxation. For the COVERAGE problem, this yields an $O(\log m)$-approximation algorithm, which is tight, since the problem generalizes SET-COVER. This improves upon the $O(b \cdot \log n)$-approximation algorithm for the min-sum version of the problem, where b is a certain graph parameter which can be as large as $\Omega(n)$ [Algorithmica '12]. The same relaxation can also be used to get an k-approximation algorithm, where k is the number of different interfaces. This generalizes a similar result for the uniform cost case. For the CONNECTIVITY problem, we obtain an $O(\log^2 m)$-approximation algorithm, which is the first non-trivial approximation algorithm for this problem. The algorithm is based on a similar LP relaxation with additional cut constraints to ensure connectivity. The rounding procedure is similar to the one for the COVERAGE problem but requires a more careful analysis to ensure that the connectivity constraints are satisfied.

1 Introduction

Designing Internet of Things (IoT) networks often demands establishing connection between large number of devices, which can be achieved by activating communication interfaces at each of them. However, the devices present in the network may vary according to the type of interfaces they have available and the amount of resources a given interface consumes. For example some device may be able to utilize Bluetooth and Wi-Fi connections, while the others may provide Zigbee and Z-Wave interfaces. Since IoT devices are typically resource-constrained, it may be desirable to try to establish every relevant connection, while minimizing the amount of resources required to do so.

We model the above problem as a graph problem. Every device in the network is represented by a vertex with a set of labels encoding its available interfaces. Two devices can communicate if they share a common available interface and are connected by an edge, representing that they are within communication range. A connection is established between two devices whenever they

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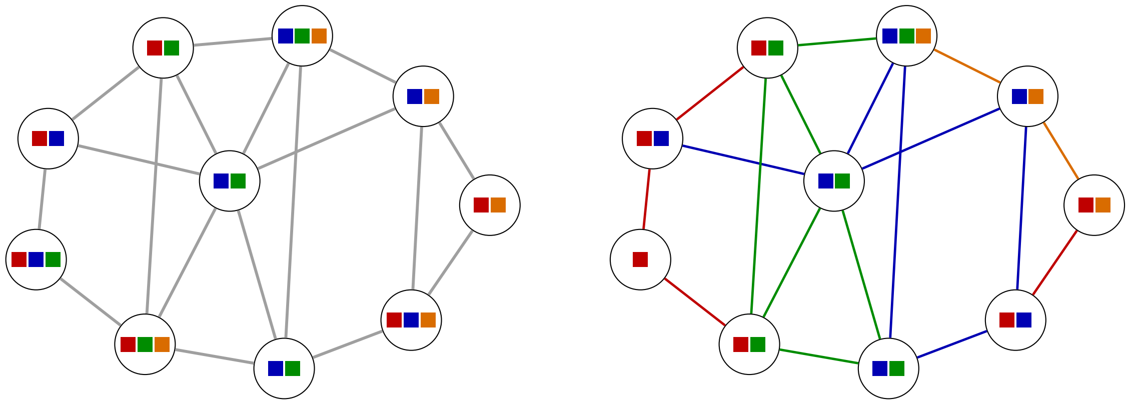
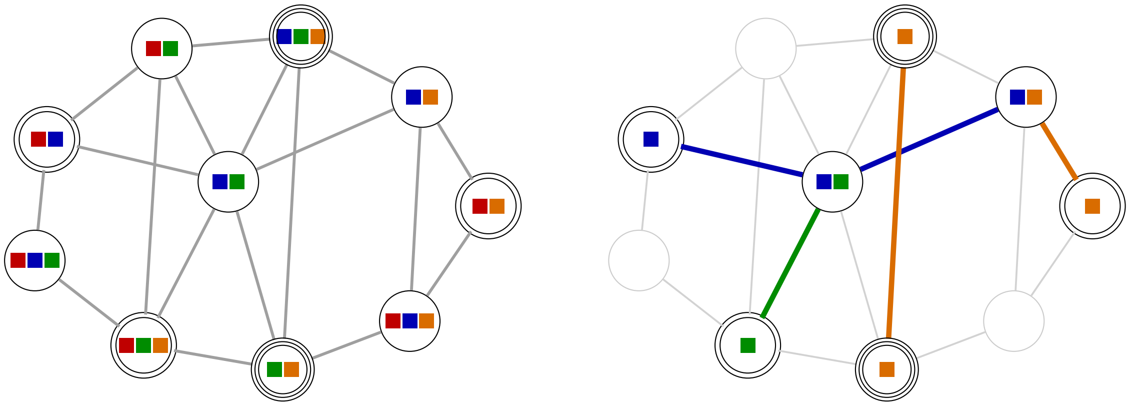


Available labels　　　　Example covering assignment

Figure 1: An example instance of the Coverage problem and a feasible covering assignment.



Available labels (double/triple contour = terminal group 1/2)　　　　Example connecting assignment

Figure 2: An example instance of the Connectivity problem and a feasible connecting assignment.

share a common active interface. We want to find an assignment of active interfaces to vertices so that we ensure either Coverage or Connectivity.

In the Coverage problem, we wish to establish every possible direct connection in the network which means that both endpoints of each edge must share a common activated interface (see Figure 1 for an example) while in the Connectivity problem, we are additionally given a collection of disjoint vertex subsets called terminals, and we require each terminal group to be (not necessarily directly) connected in the subgraph induced by active interfaces (see Figure 2 for an example). In both problems, we assume that activating an interface induces a cost that depends on the type of interface and on the vertex that activates it. The goal is to minimize the min-max cost, i. e., the maximum cost of activated interfaces across all vertices.

1.1 Related Work

In the following, let k denote the number of different interfaces across the network, n the number of vertices in the input graph, m the number of edges, Δ the maximum degree, $c_{\max}$ the maximum cost and $c_{\min}$ the minimum cost of an interface. In the *homogeneous* setting, the costs depend on the interface only while in the *heterogeneous* case they depend on the interface and on the vertex.

Coverage Problem. The algorithmic study of problems in multi-interface networks started with the problem of covering a multi-interface network, introduced in Caporuscio et al. [6]. This problem has been studied in the homogeneous setting only, both for min-sum [11] and min-max [7] objective functions.

The min-sum variant is APX-hard for any constant $k \geq 3$ and unit costs, and it is hard to approximate within a $O(\ln k)$ factor even with unit costs, but it has a $\min\{k-1, \sqrt{n}(1+\ln n)\}$-approximation [11].

The min-max variant is NP-Hard for any constant $\Delta \geq 5$, $k \geq 16$ and for unit costs, and it is not approximable within a $\eta \ln \Delta$ factor for some constant η even on trees and with unit costs. On the positive side, it admits a $(\ln \Delta + 1 + b \cdot \min\{c_{\max}, \ln \Delta + 1\})$-approximation, where b is a parameter that depends on the structural properties of the input graph and is bounded by many other graph structutal parameters such as treewidth, arboricity or maximum degree. It also has a $(1 + (k-2)\frac{c_{\max}}{2c_{\min}})$-approximation and a $(\frac{\Delta}{2}\frac{c_{\max}}{c_{\min}})$-approximation [7].

Connectivity Problem. The problem of connecting a multi-interface network was introduced by Kosowski et al. [13]. It was first studied in the homogeneous setting and in the special case where all vertices are terminals, both for the min-sum [13] and the min-max objective functions [7].

The min-sum variant where all vertices are terminals and costs are homogeneous is APX-hard for $k \geq 2$, $\Delta \geq 4$ and unit costs already, but has a 2-approximation [13]. This result was improved by Athanassopoulos et al. [4] to a randomized $(\frac{3}{2} + \epsilon)$-approximation for any constant ϵ. Moreover, this approximation algorithm applies to the more general problem where for each interface, the set of edges that it can cover is specified in the input.

The min-max variant where all vertices are terminals and costs are homogeneous is NP-Hard for any constant $\Delta \geq 3$, $k \geq 10$ and for unit costs. Similarly to the coverage problem, it is not approximable within a $\eta \ln n$ factor even for unit costs, but has simple $(1 + (k-2)\frac{c_{\max}}{2c_{\min}})$- and $\frac{\Delta}{2}\frac{c_{\max}}{c_{\min}}$-approximations [7].

Athanassopoulos et al. [4] expand the study of the min-sum variant to *group connectivity*, a more general problem where the solution graph should connect together groups of vertices that might not span the whole set of vertices. They provide a randomized 4-approximation for it. They also initiate the study of this problem in the heterogeneous setting, where costs can also depend on the vertex. They show that the min-sum variant where all vertices are terminals but costs are heterogeneous is hard to approximate within $o(\ln n)$, but the min-sum group connectivity heterogeneous case admits a $O(\ln n)$-approximation.

Other Multi-interface Problems. There has been some work on the study of the coverage problem from the perspective of parameterized complexity, possibly with an addtional constraint on the maximum number of active interfaces in a vertex [2, 1, 5], yielding efficient algorithms in some graph classes. Some papers also study other problems in multi-interface networks, like cheapest path [13], matchings [12], or flow problems [8].

1.2 Our contribution and techniques

We study the approximability of the min-max Coverage and Connectivity problems under heterogeneous cost functions. We give a set-cover-style LP relaxation for the min-max Coverage problem and show that treating the resulting fractional solution as a probability distribution and applying a randomized rounding procedure yields an $O(\log m)$-approximation. This is the first logarithmic approximation for this problem with heterogeneous costs, and it matches the previously known lower bound of $\Omega(\log n)$ for unit costs and on stars [7]. It also improves the previously best known $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$ approximation ratio [7], which was limited to the homogeneous case. A simpler deterministic rounding of the same LP gives a k-approximation, generalizing the approximation $O\Big(1 + (k-2)\frac{c_{\max}}{2c_{\min}}\Big)$ for homogeneous min-max costs[7].

For the Connectivity problem, we extend the LP by adding flow-based cut constraints encoding group connectivity. Our rounding procedure is similar to the one for the Coverage problem, however, this time our rounding is iterative and the algorithm proceeds until all terminal groups are connected. Each iteration is randomized and we show that in expectation, in each such round the algorithm covers a siginificant fraction of edges of some graph which connects all group terminals with high probability. This yields an $O(\log^2 m)$-approximation algorithm, which is the first non-trivial approximation algorithm for this problem.

The rest of the paper is structured as follows: In Section **??**, we define the problems formally, and recall the elements of probability theory useful for the analysis of our algorithms. We also describe the preprocessing used for instances of both types of the problem (coverage or connectivity). In Section **??**, we present both the $O(\log m)$-approximation as well as the k-approximation algorithms for the Coverage. In Section **??**, we present the $O(\log^2 m)$-approximation algorithm for Connectivity. We conclude with a short discussion on open questions and future work.

2 Preliminaries

Let $G = (V, E)$ be a connected simple graph. Let $[k]$ be the set of different types of interfaces across the network and let $\lambda\colon V \to 2^{[k]}$ be a function that assigns to each vertex $v \in V$ its set of available interfaces, such that for each edge $uv \in E$, $\lambda(u) \cap \lambda(v) \neq \emptyset$. An assignment of active interfaces is a function $\lambda_A\colon V \to 2^{[k]}$ such that $\lambda_A(v) \subseteq \lambda(v)$ for every $v \in V$. We say that an edge $uv \in E$ is *covered* if $\lambda_A(u) \cap \lambda_A(v) \neq \emptyset$. We denote by $\operatorname{cov}(\lambda_A) = \{uv \in E : \lambda_A(u) \cap \lambda_A(v) \neq \emptyset\}$ the set of edges covered by λ_A, and by G_A the subgraph of G with edge set $\operatorname{cov}(\lambda_A)$.

In the Coverage problem, we wish to find an assignment of active interfaces λ_A such that $G_A = G$. We call such an assignment a *covering assignment*.

In the Connectivity problem, we are additionally given $p \geq 1$ pairwise disjoint groups of *terminals* $U_1, U_2, \ldots, U_p \subseteq V$ and we wish to find an assignment of active interfaces λ_A such that for every $r \in [p]$, $U_r(G_A)$ and all vertices in U_r lie in the same connected component of G_A. Vertices in $V \setminus \bigcup_{r=1}^{p} U_r$ may be used as Steiner vertices. We call such an assignment a *connecting assignment*. The special case $p = 1$ with $U_1 = V$ corresponds to requiring that G_A is a connected spanning subgraph of G.

We are also given a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$, which denotes the cost of activating the interface $i \in [k]$ at the vertex $v \in V$. For an assignment λ_A, define the *cost* of v as:

$$\operatorname{cost}(v, \lambda_A) = \sum_{i \in \lambda_A(v)} c(i, v)\,.$$

Based on this, we consider two global cost criteria. The *max-cost* of an assignment λ_A is:

$$\operatorname{cost}_M(\lambda_A) = \max_{v \in V} \operatorname{cost}(v, \lambda_A)$$

and the *sum-cost* of an assignment λ_A is:

$$\operatorname{cost}_S(\lambda_A) = \sum_{v \in V} \operatorname{cost}(v, \lambda_A)\,.$$

The goal of the min-max (resp. min-sum) assignment problem is to find an assignment λ_A that minimizes cost_M (resp. cost_S). Throughout this paper, we focus on the *min-max* cost criterion. We will simply write $\operatorname{cost}(\lambda_A)$ instead of $\operatorname{cost}_M(\lambda_A)$ and we will refer to the min-max assignment problem as the Coverage and Connectivity problems, respectively.

2.1 Probabilistic tools

For the sake of our analysis we will need the following probability facts:

Theorem 2.1 (Markov's Inequality)**.** *Let X be a non-negative random variable and $a > 0$. Then:*

$$\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}.$$

Theorem 2.2 (Wald's Identity [15])**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables with finite expectation $\mathbb{E}[X]$, and let N be a non-negative integer-valued random variable independent of the sequence $(X_i)_{i\geq 1}$ with finite expectation $\mathbb{E}[N]$. Then:*

$$\mathbb{E}\left[\sum_{i=1}^{N} X_i\right] = \mathbb{E}[N] \cdot \mathbb{E}[X].$$

Theorem 2.3 ((Weighted) Chernoff Bound)**.** *Let $X_1, X_2, \ldots, X_n$ be independent random variables taking values in $[0, 1]$, let $w_1, w_2, \ldots, w_n$ be non-negative weights, such that for every $i \in [n]$ we have that $w_i \in [0, 1]$. Let $X = \sum_{i=1}^{n} w_i \cdot X_i$ and let $\mu = \mathbb{E}[X]$. Then for every $\delta_1 > 0$ and $\delta_2 \in (0, 1)$ we have that:*

$$\Pr(X \geq (1 + \delta_1) \cdot \mu) \leq \left(\frac{e^{\delta_1}}{(1 + \delta_1)^{1+\delta_1}}\right)^{\mu} \qquad \Pr(X \leq (1 - \delta_2) \cdot \mu) \leq \left(\frac{e^{-\delta_2}}{(1 - \delta_2)^{1-\delta_2}}\right)^{\mu}.$$

The proof of the above fact follows the proof of the standard Chernoff bound and we include it in the appendix for completeness.

2.2 Preprocessing

Scaling of costs Due to technical reasons, we need to preprocess the input in order to produce a polynomial number of instances, in such a way that for at least one of them we have a good lower bound on OPT. We will also allow the costs to be rational numbers. By testing consecutive powers of 2, we guess the smallest number b, such that the costliest activated interface in the optimal solution has cost at most 2^b (and the instance still has a solution). Note that the number of guesses is polynomial. For each vertex $v \in V$, we discard all of the interfaces from $\lambda(v)$ with cost larger than 2^b and we normalize the costs of remaining interfaces by dividing them by 2^b, so that the largest cost is exactly 1. We get that $\text{OPT} \geq 1/2$ (for the right choice of b).

Cheap and expensive vertices Having established the scaling of costs, we now consider two types of vertices. For a given vertex $v \in V$, if $\sum_{i\in\lambda(v)} c(i, v) \leq 1$, then we can safely assume that all interfaces in $\lambda(v)$ are activated at v, since the cost of activating all of them is at most 1 and thus at most $2 \cdot \text{OPT}$. We call such vertices *cheap*. The remaining vertices are called *expensive*. Note that for every expensive vertex v, we can assume that $\text{cost}(\lambda_A, v) \geq 1$, since such an assumption will always result in an increase of cost of at most $2 \cdot \text{OPT}$ which does not impact the asymptotical guarantee of our procedures.

3 Approximation algorithms for the coverage problem

In this section, we consider the Coverage problem with heterogeneous costs. According to the preprocessing, we assume that interface costs are in $[0, 1]$ and $\text{OPT} \geq 1/2$. The considered problem is formalized below:

Coverage

Input: A connected simple graph $G = (V, E)$, available interfaces $\lambda\colon V \to 2^{[k]}$, a cost function $c\colon [k] \times V \to [0, 1]$.
Task: Find a covering assignment λ_A minimizing $\text{cost}(\lambda_A)$.

The idea behind our ILP formulation and the algorithms are inspired by the rounding procedure known for Set Cover [14] and related problems [3].

3.1 ILP formulations of the Coverage problem

We start with the ILP formulation of min-max Coverage. Let $x_{i,v}$ be a binary variable that is 1 iff the interface $i \in \lambda(v)$ is assigned to vertex $v \in V$. If a given vertex v is a cheap, then without large cost increase we can force $x_{i,v} = 1$, for every $i \in \lambda(v)$. Otherwise, if a vertex v is expensive, we can assume that $\sum_{i\in\lambda(v)} x_{i,v} \cdot c(i, v) \geq 1$, which corresponds to the fact that $\text{cost}(\lambda_A, v) \geq 1$. Thus, we can write the following ILP formulation of Coverage:

$$\begin{aligned}
\min \quad & M \\
\text{s.t.} \quad & \sum_{i\in\lambda(v)} c(i,v)\cdot x_{i,v} \leq M \quad \forall v \in V && (1)\\
& \sum_{i\in\lambda(u)\cap\lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq 1 \quad \forall uv \in E && (2)\\
& \begin{cases} x_{i,v} = 1 \quad \forall i \in \lambda(v) & \text{if } \sum_{i\in\lambda(v)} c(i,v)\cdot x_{i,v} < 1 \\ \sum_{i\in\lambda(v)} c(i,v)\cdot x_{i,v} \geq 1 & \text{otherwise} \end{cases} \quad \forall v \in V && (3)\\
& x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v) && (4)
\end{aligned}$$

Constraints of type (**??**) enforce that the cost of each vertex does not exceed the global cost M. Constraints (**??**) enforce that for every edge $uv \in E(G)$, there exists a common activated interface. Constraints (**??**) enforce that for every cheap vertex, all of its interfaces are activated, while for every expensive vertex, the cost of activated interfaces is at least 1. Finally, constraints (**??**) enforce the integrality of the solution. Note that the above formulation is not linear due to constraints (**??**). However, by adding an additional variables $z_{i,uv}$, we can linearize constraints (**??**) by replacing them with the constraints $\sum_{i\in\lambda(u)\cap\lambda(v)} z_{i,uv} \geq 1$ together with $z_{i,uv} \leq x_{i,u}$ and $z_{i,uv} \leq x_{i,v}$ for every $i \in \lambda(u) \cap \lambda(v)$ and $uv \in E$. Thus, we can solve the LP relaxation of this formulation in polynomial time.

3.2 A $O(\log m)$-approximation algorithm for the Coverage problem

In order to extract an integral solution out of the LP-relaxation, we employ a randomized procedure. For each type of an interface we draw a uniformly random threshold and activate the interface at a vertex if the corresponding variable (scaled up by a factor of $O(\log m)$) in the fractional solution is above the threshold.

Algorithm 1: $O(\log m)$-approximation algorithm for the Coverage problem

1 Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i\in\lambda(v), v\in V}$.
2 Pick independent uniformly random thresholds $t_i \in (0, 1)$, for each $i \in [k]$.
3 $\lambda_A(v) \leftarrow \{i \in \lambda(v)\colon 3 \ln m \cdot x_{i,v} \geq t_i\}$, for every $v \in V$.
4 **return** $\{\lambda_A(v)\}_{v\in V}$.

Theorem 3.1. *Algorithm* **??** *is an $O(\log m)$-approximation algorithm for* min-max Coverage *which succeeds with high probability.*

Proof. Firstly, we claim that the algorithm indeed returns a feasible solution. We have the following lemma:

Lemma 3.2. *With high probability, Algorithm* **??** *returns a covering assignment.*

Proof. Consider an edge $uv \in E$. We have:

$$\begin{aligned}
\Pr[\lambda_A(u) \cap \lambda_A(v) = \emptyset] &= \prod_{i \in \lambda(u) \cap \lambda(v)} \Pr[t_i > 3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}] \\
&= \prod_{i \in \lambda(u) \cap \lambda(v)} \{1 - 3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}\} \\
&\leq \prod_{i \in \lambda(u) \cap \lambda(v)} \exp\{-3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}\} \\
&= \exp\left\{-3 \ln m \cdot \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\}\right\} \\
&\leq \exp\{-3 \ln m\} = \frac{1}{m^3}
\end{aligned}$$

By applying the union bound over all m edges we get that the algorithm returns a covering assignment with probability at least $1 - \frac{m}{m^3} = 1 - \frac{1}{m^2}$. □

Now, we also wish to show that the approximation factor achieved by our algorithm is upper bounded by $O(\log m)$. We have the following lemma:

Lemma 3.3. *Algorithm* **??** *returns a solution of cost at most* $9 \ln m \cdot \mathrm{OPT}$ *with high probability.*

Proof. Fix a vertex $v \in V$. We want to show that with high probability, the cost of activating interfaces at v is at most $9 \ln m$ times the cost of activating interfaces at v in the fractional solution. If v is cheap, then the cost of activating all interfaces at v is at most 1, so the claim holds. Assume that v is expensive. By $X_{v,i}$ denote the random variable that is 1 if $3 \ln m \cdot x_{i,v} \geq t_i$ and 0 otherwise. Then we can write $\mathrm{cost}(v, \lambda_A) = \sum_{i \in \lambda(v)} X_{v,i} \cdot c(i, v)$. We have: $\mathbb{E}[X_{v,i}] = \Pr[i \in \lambda_A(v)] \leq 3 \ln m \cdot x_{i,v}$. Thus, $\mathbb{E}[\mathrm{cost}(v, \lambda_A)] \leq 3 \ln m \cdot \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i, v)$. By applying the Chernoff bound with $\delta = 2$ we get that

$$\Pr[\mathrm{cost}(v, \lambda_A) \geq 3 \cdot \mathbb{E}[\mathrm{cost}(v, \lambda_A)]] \leq \left(\frac{e^2}{27}\right)^{\mathbb{E}[\mathrm{cost}(v, \lambda_A)]} \leq \exp\{-\mathbb{E}[\mathrm{cost}(v, \lambda_A)]\} \leq \frac{1}{m^3}$$

where the last inequality is by using the fact that, $\sum_{i \in \lambda(v)} x_{i,v} \cdot c(i, v) \geq 1$. By applying the union bound over all n vertices we get that with probability at least $1 - \frac{n}{m^3} \geq 1 - \frac{1}{m}$ (assuming $m \geq 2$) for every $v \in V$, $\mathrm{cost}(v, \lambda_A) \leq 9 \ln m \cdot \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i, v)$. Thus, with high probability we have $\mathrm{cost}(\lambda_A) \leq 9 \ln m \cdot \max_{v \in V} \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i, v) \leq 9 \ln m \cdot \mathrm{OPT}$. □

Combining the above lemmas gives the desired result. □

3.3 A k-approximation algorithm

In this section we show that the LP solution can also be used to obtain a k-approximation. Note that for this algorithm we do not need the assumption that $\mathrm{OPT} \geq 1/2$, so it also can be adopted to work for the min-sum variant of the problem (assuming the cost criterion of the ILP is appropriately changed).

Algorithm 2: k-approximation algorithm

1 Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i \in \lambda(v), v \in V}$.
2 Set $\lambda_A(v) \leftarrow \{i \in \lambda(v) \colon x_{i,v} \geq \frac{1}{k}\}$, for every $v \in V$.
3 **return** $\{\lambda_A(v)\}_{v \in V}$.

Theorem 3.4. *Algorithm **??** is a k-approximation algorithm for the* Coverage *problem.*

Proof. Consider an edge $uv \in E$. By averaging, there exists an interface $i \in \lambda(u) \cap \lambda(v)$ such that $\min\{x_{i,u}, x_{i,v}\} \geq \frac{1}{k}$. Thus, $i \in \lambda_A(u) \cap \lambda_A(v)$ so λ_A is a covering assignment. Moreover, since $x_{i,v} \geq \frac{1}{k}$ for every $i \in \lambda_A(v)$, we have that $c(\lambda_A(v)) = \sum_{i \in \lambda_A(v)} c(i,v) \leq k \cdot \sum_{i \in \lambda_A(v)} c(i,v) \cdot x_{i,v}$. Thus, we have $\mathrm{cost}(\lambda_A) \leq k \cdot \max_{v \in V} \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} \leq k \cdot \mathrm{OPT}$. □

4 Approximation algorithm for the Connectivity problem

In this section we consider the Connectivity problem with heterogeneous costs. We again assume that the instance is preprocessed, so we have the assumption that $\mathrm{OPT} \geq 1/2$.

Connectivity

Input: A connected simple graph $G = (V, E)$, available interfaces $\lambda \colon V \to 2^{[k]}$, a cost function $c \colon [k] \times V \to [0,1]$, pairwise disjoint groups of terminals $U_1, \ldots, U_p \subseteq V$.
Task: Find a connecting assignment λ_A minimizing $\mathrm{cost}(\lambda_A)$.

4.1 ILP formulation of the Connectivity problem

We use a similar LP to model the Connectivity problem, however we use additional flow variables to encode the connectivity requirement. The $x_{i,v}$ variables are defined as before. For each edge e, we add a variable y_e which is 1 iff the edge e is covered by the solution. For any $S \subseteq V$, let $\delta(S)$ denote the set of edges with exactly one endpoint in S. Connectivity within each group U_r is enforced via cut constraints: for every $r \in [p]$ and every cut $(S, V \setminus S)$ that separates at least one pair of terminals in U_r, the sum of y-variables of edges crossing the cut, $y(\delta(S)) = \sum_{e \in \delta(S)} y_e \geq 1$. We can formulate Connectivity as follows:

$$
\begin{aligned}
\min \quad & M \\
\text{s.t.} \quad & \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) \leq M \quad \forall v \in V && (5) \\
& \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq y_{uv} \quad \forall uv \in E && (6) \\
& \begin{cases} x_{i,v} = 1 \quad \forall i \in \lambda(v) & \text{if } \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} < 1 \\ \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} \geq 1 & \text{otherwise} \end{cases} \quad \forall v \in V && (7) \\
& y(\delta(S)) \geq 1 \quad \forall r \in [p],\ \forall S \subset V \colon S \cap U_r \neq \emptyset,\ (V \setminus S) \cap U_r \neq \emptyset && (8) \\
& x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v) \\
& y_{uv} \in \{0,1\} \quad \forall uv \in E && (9)
\end{aligned}
$$

Observe that the above ILP has an exponential number of constraints, but we can solve its LP relaxation in polynomial time by using the ellipsoid method and a separation oracle for the cut constraints, which can be implemented using a max-flow algorithm.

4.2 An $O(\log^2 m)$-approximation algorithm for the Connectivity problem

Obtaining an integral solution for Connectivity requires a more careful rounding procedure than for Coverage. We wish to ensure that at least one subgraph which spans all terminal groups is covered by the solution returned by the algorithm. To do so, we start by randomly sampling edges according to the y-variables of the fractional solution, so that with high probability, the sampled subgraph H contains a spanning subgraph for each terminal group. To achieve this goal, we set the probability of choosing e to be $p_e = \min\{1, 4 \ln m \cdot y_e\}$. Then, the goal of the algorithm becomes covering H, so we repeat the following process until this is the case: we draw uniformly random thresholds t_i for each type of interface $i \in [k]$, and activate i, at any vertex for which $x_{i,v} \geq t_i$. Intuitevely, each iteration of this process costs $O(\mathrm{OPT})$ in expectation and covers $\Omega(1/\log m)$ fraction of edges of H. The additional $O(\log m)$ factor in the approximation ratio comes from the fact that in order to ensure connectivity of H, we scaled up the probabilities of choosing edges by a factor of $O(\log m)$. This forces the expected number of iterations to be of order $\Omega\big(\log^2 m\big)$.

Algorithm 3: $O\big(\log^2 m$-approximation algorithm for the CONNECTIVITY problem

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Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i\in\lambda(v),v\in V}$, $\{y_{uv}\}_{uv\in E}$.
For every edge $e \in E$, $p_e \leftarrow \min\{1, 4\ln m \cdot y_e\}$.
With probability p_e set $Y_e \leftarrow 1$, otherwise set $Y_e \leftarrow 0$.
Let H be the subgraph of G induced by the edges e such that $Y_e = 1$.
$\mathcal{S} \leftarrow \emptyset$.
$j \leftarrow 1$.
$\lambda_A \leftarrow \{\emptyset\}_{v\in V}$.
while $\mathrm{cov}(\lambda_A, H) \neq E(H)$ **do**
    Pick independent uniformly random thresholds $t_i \in (0,1)$, for each $i \in [k]$.
    $\lambda_A^j(v) \leftarrow \{i \in \lambda(v) \colon x_{i,v} \geq t_i\}$, for every $v \in V$.
    $\lambda_A \leftarrow \lambda_A^j \cup \lambda_A$.
    $j \leftarrow j + 1$.
return λ_A

Theorem 4.1. *Algorithm ?? is an $O\big(\log^2 m\big)$-approximation algorithm for* MIN-MAX CONNECTIVITY *which succeeds with high probability.*

Proof. In order to show that the algorithm returns a feasible solution, we first show that with high probability, the subgraph H connects all terminals in each group U_r. It should be remarked that constructing explicitly H is not necessary for the algorithm to work, nonetheless, we need to show that the solution returned by the algorithm is feasible. We start with the following lemma:

Lemma 4.2. *With high probability, for every $r \in [p]$, H connects all terminals in U_r.*

Proof. Fix any $r \in [p]$ and any $S \subset V$ with $S \cap U_r \neq \emptyset$ and $(V \setminus S) \cap U_r \neq \emptyset$. Let $\ell \in \mathbb{N}$, such that $\ell \leq y(\delta(S)) < \ell + 1$. We have:

$$\begin{aligned}
\Pr[|\delta_H(S)| = 0] &= \prod_{e\in\delta(S)} \Pr[Y_e = 0] = \prod_{e\in\delta(S)} (1 - p_e) \\
&\leq \prod_{e\in\delta(S)} \exp\{-p_e\} \leq \exp\left\{ - \sum_{e\in\delta(S)} \min\{1,\, 4\ln m \cdot y_e\} \right\} \\
&\leq \exp\{-4\ell \ln m\} = \frac{1}{m^{4\ell}}
\end{aligned}$$

where the last inequality uses the LP constraint $y(\delta(S)) \geq 1$ for every U_r-separating cut, so $\sum_{e\in\delta(S)} y_e \geq \ell$.

A cut in G is called an α-minimum cut if its value is less than α times the value of a minimum cut in G. We now make a use of the following Karger's lemma [10, 9]:

Lemma 4.3 (Karger's lemma). *The number of α-minimum cuts in a graph is at most $n^{2\alpha}$.*

Observe that by construction of the LP, every U_r-separating fractional cut has value at least 1, so there are at most $n^{2\ell}$ cuts that are U_r-separating and have fractional value less than ℓ. Fix some $r \in [p]$. Applying the union bound over all all U_r-separating cuts we get:

$$\Pr\big[\exists\, U_r\text{-sep. cut } S\colon |\delta_H(S)| = 0\big] \le \sum_{\ell=1}^{\infty} \sum_{\substack{S \subset V,\, S \cap U_r \neq \emptyset \\ (V \setminus S) \cap U_r \neq \emptyset,\, y(\delta(S)) < \ell+1}} \Pr[|\delta_H(S)| = 0]$$

$$\le \sum_{\ell=1}^{\infty} n^{2\ell} \cdot \frac{1}{m^{4\ell}} = \sum_{\ell=1}^{\infty} \frac{1}{m^{2\ell}} = \frac{1}{m^2 - 1}$$

Thus, applying the union bound over all groups $r \in [p]$, no U_r-separating cut is empty in H for any $r \in [p]$ with probability at least $1 - \frac{p}{m^2-1}$. Since $p \le n \le m$, we get that with high probability, for every $r \in [p]$, the subgraph H connects all terminals in U_r with high probability. $\square$

Having shown that with high probability, the subgraph H connects all terminals in each group U_r, we now wish to show that the assignment λ_A returned by the algorithm covers all of the edges of H. Let j be an index of some iteration of the while loop. Let H_j be the subgraph of H that is not covered by λ_A at the beginning of the j-th iteration of the while loop. We wish to show, that in expectation, the solution λ_A produced in the j-th iteration of the while loop, call it λ_A^j, covers a significant fraction of the edges in H_j. We have the following lemma:

Lemma 4.4. *We have* $\mathbb{E}\left[\left|\mathrm{cov}\left(\lambda_A^j, H_j\right)\right|\right] \ge \frac{1-1/e}{3 \ln m} \cdot \mathbb{E}\left[|E(H_j)|\right]$.

Proof. Consider any edge $uv \in E(G)$. Observe that $\mathbb{E}[Y_{uv}] \le 4 \ln m \cdot y_{uv}$. By the definition of the LP we get that:

$$\sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \ge y_{uv} \ge \frac{\mathbb{E}[Y_{uv}]}{4 \ln m}.$$

Therefore, the probability that uv is uncovered by λ_A is at most:

$$\begin{aligned}
\Pr[\lambda_A(u) \cap \lambda_A(v) = \emptyset] &= \prod_{i \in \lambda(u) \cap \lambda(v)} \Pr[t_i > \min\{x_{i,u}, x_{i,v}\}] \\
&= \prod_{i \in \lambda(u) \cap \lambda(v)} \{1 - \min\{x_{i,u}, x_{i,v}\}\} \\
&\le \prod_{i \in \lambda(u) \cap \lambda(v)} \exp\{-\min\{x_{i,u}, x_{i,v}\}\} \\
&= \exp\left\{-\sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\}\right\} \\
&\le \exp\left\{-\frac{\mathbb{E}[Y_{uv}]}{4 \ln m}\right\}
\end{aligned}$$

Thus, we have $\Pr[\lambda_A(u) \cap \lambda_A(v) \neq \emptyset] \ge 1 - \exp\left\{-\frac{\mathbb{E}[Y_{uv}]}{4 \ln m}\right\}$. By summing up over all edges in H_j we lower bound the expected number of edges covered by λ_A^j as follows:

$$\begin{aligned}
\mathbb{E}\left[\left|\mathrm{cov}\left(\lambda_A^j, H_j\right)\right|\right] &= \sum_{uv \in E(H_j)} \Pr[\lambda_A(u) \cap \lambda_A(v) \neq \emptyset] \\
&\ge \sum_{uv \in E(H_j)} \left\{1 - \exp\left\{-\frac{\mathbb{E}[Y_{uv}]}{4 \ln m}\right\}\right\} \\
&\ge (1 - 1/e) \cdot \sum_{uv \in E(H_j)} \frac{\mathbb{E}[Y_{uv}]}{4 \ln m} \ge \frac{1 - 1/e}{4 \ln m} \cdot \mathbb{E}\left[|E(H_j)|\right]
\end{aligned}$$

where the second ineqaulity is using the fact $1-e^x \le (1-1/e)\cdot x$, for $x \in [0,1]$ and last inequality is by the linearity of expectation. $\square$

Now, we want to show that the number of iterations of the while loop is at most $O\big(\log^2 m\big)$ with high probability. Let I be the random variable denoting the number of these iterations. We firstly bound the expected value of I.

Lemma 4.5. $\mathbb{E}[I] \le \frac{9\ln^2 m}{1-1/e}$.

Proof. Let $\alpha = \frac{1-1/e}{4\ln m}$. By definition, $|E(H_{j+1})| = |E(H_j)| - \left|\mathrm{cov}\Big(\lambda_A^j, H_j\Big)\right|$. By linearity of expectation and the previous lemma, $\mathbb{E}\left[|E(H_{j+1})|\right] \le (1-\alpha)\cdot \mathbb{E}\left[|E(H_j)|\right]$. Unfolding the recurrence we get that, $\mathbb{E}\left[|E(H_j)|\right] \le (1-\alpha)^{j-1}\cdot m \le me^{-\alpha(j-1)}$. By Markov's inequality, $\Pr\{|E(H_j)| \ge 1\} \le me^{-\alpha(j-1)}$. Let $j^* = \left\lfloor \frac{9\ln^2 m}{1-1/e} \right\rfloor$. We have:

$$\mathbb{E}[I] = \sum_{j=1}^{\infty} \Pr[I \ge j] = \sum_{j=1}^{j^*} \Pr[I \ge j] + \sum_{j=j^*+1}^{\infty} \Pr[I \ge j]$$
$$\le j^* + \sum_{j=j^*+1}^{\infty} me^{-\alpha(j-1)} = j^* + m\cdot \frac{e^{-\alpha j^*}}{1-e^{-\alpha}}.$$

Since $j^* \ge \frac{2\ln m}{\alpha} = \frac{8\ln^2 m}{1-1/e}$, we have $me^{-\alpha j^*} \le me^{-2\ln m} = \frac{1}{m}$, and $\frac{1}{1-e^{-\alpha}} \le \frac{2}{\alpha} = \frac{8\ln m}{1-1/e}$ for $m \ge 2$. Therefore, $\mathbb{E}[I] \le j^* + \frac{1}{m}\cdot\frac{8\ln m}{1-1/e} \le \frac{9\ln^2 m}{1-1/e}$ for high enough m. $\square$

Lemma 4.6. $\Pr\left[I \ge \frac{9\ln^2 m}{1-1/e}\right] \le \frac{1}{m}$.

Proof. Let $\alpha = \frac{1-1/e}{4\ln m}$ and $j^* = \frac{9\ln^2 m}{1-1/e}$. As shown in the proof of Lemma 4.5, $\Pr\{|E(H_j)| \ge 1\} \le me^{-\alpha j}$. Setting $j = j^* \ge \frac{2\ln m}{\alpha} = \frac{8\ln^2 m}{1-1/e}$ gives $me^{-\alpha j^*} \le me^{-2\ln m} = \frac{1}{m}$. Hence, $\Pr\left[I \ge j^*\right] = \Pr\{|E(H_{j^*})| \ge 1\} \le \frac{1}{m}$. $\square$

We get that the procedure executes at most $O\big(\log^2 m\big)$ iterations with high probability, so the running time of the algorithm is polynomial. It remains to show that the cost of the solution returned by the algorithm is at most $O\big(\log^2 m\big)\cdot \mathrm{OPT}$. We have:

Lemma 4.7. *With high probability, we have that* $\mathrm{cost}(\lambda_A) \le \frac{27\ln^2 m}{1-1/e}\cdot \mathrm{OPT}$.

Proof. Fix an expensive vertex $v \in V$, cheap vertices have cost at most $1 \le 2\cdot\mathrm{OPT}$ regardless of I. Let $\mu_v = \sum_{i\in\lambda(v)} x_{i,v}\cdot c(i,v) \ge 1$.

By $X_{v,i}^j$ denote the indicator that $x_{i,v} \ge t_i$ in iteration j. Since different iterations draw independent fresh thresholds, the cost variables $\left\{\mathrm{cost}\Big(v, \lambda_A^j\Big)\right\}_{j\ge1}$ are i.i.d. with mean μ_v, and I is independent of the sequence $\left\{\mathrm{cost}\Big(v, \lambda_A^j\Big)\right\}_{j\ge1}$. Applying Wald's identity:

$$\mathbb{E}\left[\mathrm{cost}(v,\lambda_A)\right] = \mathbb{E}[I]\cdot\mu_v \le \frac{9\ln^2 m}{1-1/e}\cdot\mu_v.$$

We now use the weighted Chernoff bound with $\delta = 2$:

$$\Pr\left[\mathrm{cost}(v,\lambda_A) \ge \frac{27\ln^2 m}{1-1/e}\cdot\mu_v\right] \le \left(\frac{e^2}{27}\right)^{\frac{9\ln^2 m}{1-1/e}\cdot\mu_v} \le \exp\left\{-\frac{9\ln^2 m}{1-1/e}\cdot\mu_v\right\} \le \frac{1}{m^9},$$

where the last inequality uses $\frac{9\ln^2 m}{1-1/e} \geq 9\ln m$ for $m \geq 2$ and $\mu_v \geq 1$. By the union bound over all $n \leq m$ expensive vertices, with probability at least $1 - \frac{1}{m^8}$, for every $v \in V$:

$$\mathrm{cost}(v, \lambda_A) \leq \frac{27\ln^2 m}{1-1/e} \cdot \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) \leq \frac{27\ln^2 m}{1-1/e} \cdot \mathrm{OPT}.$$

□

We conclude, that with high probability the algorithm returns a solution which covers all edges in H in at most $O(\log^2 m)$ iterations and has cost at most $O(\log^2 m) \cdot \mathrm{OPT}$. Since H connects every group U_r internally, the solution returned by the algorithm is a valid solution to the CONNECTIVITY problem with cost at most $O(\log^2 m) \cdot \mathrm{OPT}$. □

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A Weighted Chernoff Bound Derivation

Theorem A.1 ((Weighted) Chernoff bound)**.** *Let $X_1, X_2, \ldots, X_n$ be independent random variables taking values in $[0,1]$, let $w_1, w_2, \ldots, w_n$ be non-negative weights, such that for every $i \in [n]$ we have that $w_i \in [0,1]$. Let $X = \sum_{i=1}^{n} w_i \cdot X_i$ and let $\mu = \mathbb{E}[X]$. Then for every $\delta_1 > 0$ and $\delta_2 \in (0,1)$ we have that:*

$$\Pr(X \geq (1+\delta_1)\cdot\mu) \leq \left(\frac{e^{\delta_1}}{(1+\delta_1)^{1+\delta_1}}\right)^{\mu} \qquad \Pr(X \leq (1-\delta_2)\cdot\mu) \leq \left(\frac{e^{-\delta_2}}{(1-\delta_2)^{1-\delta_2}}\right)^{\mu}.$$

Proof. We first prove the upper-tail bound for $\delta_1 > 0$. For any $\lambda > 0$, Markov's inequality gives:

$$\Pr(X \geq (1+\delta_1)\cdot\mu) = \Pr\Big(e^{\lambda X} \geq e^{\lambda\cdot(1+\delta_1)\cdot\mu}\Big) \leq \frac{\mathbb{E}\big[e^{\lambda X}\big]}{e^{\lambda\cdot(1+\delta_1)\cdot\mu}}.$$

By independence, we have $\mathbb{E}\big[e^{\lambda\cdot X}\big] = \prod_{i=1}^{n}\mathbb{E}\big[e^{\lambda\cdot w_i\cdot X_i}\big]$. Fix $i \in [n]$. Since $w_i \cdot X_i \in [0,1]$, by convexity of $y \mapsto e^{\lambda\cdot y}$ on $[0,1]$, for every $y \in [0,1]$, $e^{\lambda\cdot y} \leq (1-y)\cdot e^0 + y \cdot e^{\lambda} = 1 + \big(e^{\lambda}-1\big)\cdot y$. Substituting $y = w_i \cdot X_i$, taking expectation, and using $1 + t \leq e^t$, we obtain:

$$\mathbb{E}\Big[e^{\lambda\cdot w_i\cdot X_i}\Big] \leq 1 + \Big(e^{\lambda}-1\Big)\cdot w_i \cdot \mathbb{E}[X_i] \leq \exp\Big(\Big(e^{\lambda}-1\Big)\cdot w_i\cdot\mathbb{E}[X_i]\Big).$$

Hence:

$$\mathbb{E}\Big[e^{\lambda X}\Big] \leq \exp\left(\Big(e^{\lambda}-1\Big)\cdot\sum_{i=1}^{n} w_i\cdot\mathbb{E}[X_i]\right) = \exp\Big(\Big(e^{\lambda}-1\Big)\cdot\mu\Big).$$

Therefore, $\Pr(X \geq (1+\delta_1)\cdot\mu) \leq \exp\big(\mu\cdot\big(e^{\lambda}-1-\lambda\cdot(1+\delta_1)\big)\big)$. The function $f(\lambda) = e^{\lambda} - 1 - \lambda\cdot(1+\delta_1)$ is minimized at $\lambda = \ln(1+\delta_1)$, and so:

$$\Pr(X \geq (1+\delta_1)\cdot\mu) \leq \exp(-\mu\cdot((1+\delta_1)\cdot\ln(1+\delta_1)-\delta_1)) = \left(\frac{e^{\delta_1}}{(1+\delta_1)^{1+\delta_1}}\right)^{\mu}.$$

Now let $\delta_2 \in (0,1)$. For any $t > 0$, again by Markov's inequality:

$$\Pr(X \le (1-\delta_2)\cdot\mu) = \Pr\left(e^{-tX} \ge e^{-t\cdot(1-\delta_2)\cdot\mu}\right) \le \frac{\mathbb{E}\left[e^{-t\cdot X}\right]}{e^{-t\cdot(1-\delta_2)\cdot\mu}}.$$

Applying the same mgf estimate with $\lambda = -t < 0$ yields $\mathbb{E}\left[e^{-t\cdot X}\right] \le \exp\left(\left(e^{-t}-1\right)\cdot\mu\right)$, so:

$$\Pr(X \le (1-\delta_2)\cdot\mu) \le \exp\left(\mu\cdot\left(e^{-t}-1+t\cdot(1-\delta_2)\right)\right).$$

The right-hand side is minimized at $t = -\ln(1-\delta_2)$, which gives:

$$\Pr(X \le (1-\delta_2)\cdot\mu) \le \exp(-\mu\cdot(\delta_2+(1-\delta_2)\cdot\ln(1-\delta_2))) = \left(\frac{e^{-\delta_2}}{(1-\delta_2)^{1-\delta_2}}\right)^{\mu}.$$

$\square$